Functional Genomics in Insect Cell Models to Study Cellular Pathways

Understanding how genes shape cell behavior is essential for advancing basic biology. This project aimed to develop a functional genomics platform for studying gene function and cellular responses in insect cell models. We combined two complementary approaches: a pro-siRNA library screening workflow for identifying phenotypes associated with changes in cell viability and an eGFP reporter assay for evaluating RNA delivery and sequence-specific gene silencing.

First, we developed a modified screening workflow based on pro-siRNA technology, which provides a scalable method for generating RNA molecules that can reduce the activity of selected genes. More than thirty 96-well pro-siRNA library plates were prepared, and selected RNA samples were introduced into insect cells and analyzed using automated fluorescence imaging. At 72 hours after transfection, fluorescent markers were used to distinguish dead cells, viable cells, and cell nuclei, and the resulting images were quantified by high-content screening. The screen identified several candidate wells with elevated dead-to-live cell count ratios, indicating changes in cell viability.

Second, we developed a fluorescent reporter system to more directly evaluate RNA delivery and gene silencing. Two transfection reagents, PEI and Lipofectamine 3000, were compared for delivering messenger RNA encoding enhanced green fluorescent protein (eGFP). Flow cytometry showed that approximately 30% of cells were GFP-positive at both 24 and 48 hours following Lipofectamine 3000-mediated transfection. In a co-transfection experiment, cells received eGFP mRNA together with either an eGFP-targeting siRNA or a non-targeting control. Both high-content imaging and flow cytometry showed a lower GFP signal in the targeting-siRNA condition, providing a measurable readout of sequence-specific gene silencing.

Together, these complementary workflows enabled observation and quantification of changes in cell viability, RNA delivery, and sequence-specific reporter suppression in insect cell models.